

PROJECT DOCUMENTATION

PERFORMANCE TESTING

EduGenie – Google Gemini Powered Learning Assistant

1. Testing Objective

Performance testing verifies that the EduGenie prototype responds correctly to common application actions and that the core web and Gemini workflow operates as expected.

2. Test Environment

The documented software stack includes Python 3.x, FastAPI, Uvicorn, Google Gemini / Google GenAI Python SDK, Jinja2, HTML5, CSS3, python-dotenv, and a modern web browser.

3. Test Cases

Open application — Expected: EduGenie page loads — Status: Passed.

Ask question “Which is the largest ocean?” — Expected: concise AI answer displayed — Status: Passed.

Empty question — Expected: browser validation prevents submission — Status: Passed.

CSS loading — Expected: styles applied — Status: Passed.

Gemini connection — Expected: AI response returned — Status: Passed.

Server — Expected: application starts on configured port — Status: Passed.

4. Response Behavior

The implemented system is documented as returning a concise answer for the example question. Response time depends on network and service availability.

5. Performance Limitations

Internet connectivity is required. API usage may have provider limits or costs. Response time depends on network and service availability.

6. Conclusion

The documented test cases passed for the core prototype workflow. More extensive load and stress testing would be required for a production-scale deployment.

Prepared for project documentation and academic demonstration.